

Weikai Lin

CONTACT INFORMATION

Ph.D. Student
University of Rochester
Wegmans Hall
Rochester, NY 14627

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<https://linwk20.github.io/>
[Google Scholar](#) | [LinkedIn](#)

RESEARCH INTERESTS

I work at the intersection of computer graphics, computer architecture, and imaging systems. My research bridges the gap between hardware and software through system-level co-optimization. Current themes:

- **Generative AI, AR/VR rendering and acceleration:** neural rendering algorithms for AR/VR and their hardware accelerators. [ECCV'26, CVPR'26, SIGGRAPH Asia'25, ASPLOS'25a, ASPLOS'25b, ISCA'25, HotMobile'25, TACO'24]
- **Imaging system design:** co-design of sensors and optics with downstream vision models. [DAC'26a, DAC'26b, DAC'25, WACV'25]

EDUCATION

University of Rochester, Rochester, NY, USA
Ph.D., Computer Science
M.Sc., Computer Science
Advisor: [Yuhao Zhu](#)

Aug 2023 – May 2028 (expected)
Aug 2023 – May 2025

Peking University, Beijing, China
M.Sc., Intelligence Science and Technology
Advisor: Dingsheng Luo

Sep 2020 – Jun 2023

Tsinghua University, Beijing, China
B.E., Electronic Engineering
Advisor: Li Su

Aug 2016 – Jun 2020

HONORS AND AWARDS

- 2026, ACM/IEEE DAC Young Fellow
- 2026, ACM ASPLOS Travel Grant
- 2025, ACM ASPLOS Best Paper Award
“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering” (first author)
- 2025, OpenAI Research Grant (\$5K)
- 2025, ACM ASPLOS Travel Grant
- 2022, Baosteel's Outstanding Student Award for HMT Students
- 2022, First Prize, Peking University Challenge Cup
- 2021, First Class Scholarship for Taiwan Master's Students

PROFESSIONAL POSITIONS

Meta Platforms, Inc., CA, USA
Research Scientist Intern

May 2026 – Oct 2026

Pixocial AI Lab, Bellevue, WA, USA
Applied Research Intern
Mentor: [Haoxiang Li](#)
Topics: Physics-based video diffusion and dynamic hair rendering. [\[project page\]](#)

May 2025 – Aug 2025

University of Rochester, Rochester, NY, USA
Graduate Researcher
Teaching Assistant

Aug 2023 – present
Fall 2024, Spring 2025, Fall 2025

Advanced Micro Devices, Inc. (AMD), Beijing, China
Research Intern
Mentor: Fuwei Yang.
Topics: SLAM and DNN evaluations on AMD GPUs. Participated in a [ROCm](#) release.

Jul 2022 – Feb 2023

Peking University, Beijing, China
Graduate Researcher
Teaching Assistant
Topics: Robotics, robot learning, and autonomous driving.

Sep 2020 – Jun 2023
2021 – 2022

Chinese Academy of Sciences (CAS), Beijing, China
Engineer Intern
Topics: HLS-based FPGA wireless communication system development.

2021 – 2022

Tsinghua University, Beijing, China
Undergraduate Researcher
Mentor: Li Su.
Topics: FPGA-based wireless codec system development.

Sep 2019 – Jun 2020

Cambricon Technologies Co., Ltd., Beijing, China
Research Intern
Mentor: Miao Li.
Topics: Neural network pruning.

Jul 2019 – Aug 2019

PUBLICATIONS

* denotes equal contribution.

- [Weikai Lin](#), Sheng Zhao, Ian Ross, Carl Marshall, Sushant Kondguli, Yuhao Zhu
“[LowPowAR: Power-Constrained Tone Mapping for Augmented Reality](#)”
[TVCG 2026] *IEEE Transactions on Visualization and Computer Graphics* (ISMAR Journal Track)
- [Weikai Lin](#), Haoxiang Li, Yuhao Zhu
“[ControlHair: Synergizing Physics Simulator and Video Diffusion for Controllable Dynamic Hair Rendering](#)”
[ECCV 2026] *19th European Conference on Computer Vision*
- Zijian Gu, [Weikai Lin](#), Shuang Zhou, Zihan Chen, Song Wang
“[UniMod: Enhancing Multi-Modal Medical Diagnosis through Cross-Modality and Within-Modality Alignment](#)”
[ACM MM 2026] *34th ACM International Conference on Multimedia*
- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Boloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
“[HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training](#)”
[DAC 2026] *63rd ACM/IEEE Design Automation Conference*
- Han Yan*, Yifei Zou*, [Weikai Lin](#), Yuhao Zhu, Xuan Zhang
“[Invited Paper: A Quantitative Approach to Exploring Novel Image Sensor Architecture Towards Autonomous Edge Machine Vision](#)”
[DAC 2026] *63rd ACM/IEEE Design Automation Conference*
- He Zhu*, Xiaotong Huang*, Zihan Liu, [Weikai Lin](#), Xiaohong Liu, Zhezhi He, Jingwen Leng, Minyi Guo, Yu Feng
“[SeeLe: A Unified Acceleration Framework for Real-Time Gaussian Splatting on Mobile Devices](#)”
[CVPR 2026 Highlight] *IEEE/CVF Conference on Computer Vision and Pattern Recognition*

- [Weikai Lin](#), Sushant Kondguli, Carll Marshall, Yuhao Zhu
[“PowerGS: Display-Rendering Power Co-Optimization for Neural Rendering in Power-Constrained XR Systems”](#)
[SIGGRAPH Asia 2025] *SIGGRAPH Asia Conference Papers*
- [Weikai Lin*](#), Tianrui Ma*, Adith Bloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
[“SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision”](#)
[DAC 2025] *62nd ACM/IEEE Design Automation Conference*
- [Weikai Lin*](#), Yu Feng*, Yuhao Zhu
[“MetaSapiens: Real-Time Neural Rendering with Efficiency-Aware Pruning and Accelerated Foveated Rendering”](#)
[ASPLOS 2025 Best Paper Award] *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*
- Yu Feng*, Zheng Liu*, [Weikai Lin*](#), Zihan Liu, Jingwen Leng, Minyi Guo, Zhezhi He, Jieru Zhao, Yuhao Zhu
[“StreamGrid: Streaming Point Cloud Analytics via Compulsory Splitting and Deterministic Termination”](#)
[ASPLOS 2025] *30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*
- Yu Feng, [Weikai Lin](#), Yuge Cheng, Zihan Liu, Jingwen Leng, Minyi Guo, Chen Chen, Shixuan Sun, Yuhao Zhu
[“Lumina: Real-Time Mobile Neural Rendering by Exploiting Computational Redundancy”](#)
[ISCA 2025] *52nd IEEE/ACM International Symposium on Computer Architecture*
- Adith Bloor, [Weikai Lin](#), Tianrui Ma, Yu Feng, Yuhao Zhu, Xuan Zhang
[“Private-Eye: In-Sensor Privacy Preservation Through Optical Feature Separation”](#)
[WACV 2025] *IEEE/CVF Winter Conference on Applications of Computer Vision*
- Nan Wu*, [Weikai Lin*](#), Ruizhi Cheng*, Bo Chen, Yuhao Zhu, Klara Nahrstedt, Bo Han
[“Advancing Immersive Content Delivery with Dynamic 3D Gaussian Splatting”](#)
[HotMobile 2025] *26th International Workshop on Mobile Computing Systems and Applications*
- Yu Feng*, [Weikai Lin*](#), Zihan Liu, Jingwen Leng, Minyi Guo, Han Zhao, Xiaofeng Hou, Jieru Zhao, Yuhao Zhu
[“Potamoi: Accelerating Neural Rendering via a Unified Streaming Architecture”](#)
[TACO 2024] *ACM Transactions on Architecture and Code Optimization*
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“OW3Det: Toward Open-World 3D Object Detection for Autonomous Driving”](#)
[IROS 2024] *IEEE/RSJ International Conference on Intelligent Robots and Systems*
- Wenfei Hu, [Weikai Lin](#), Hongyu Fang, Yi Wang, Dingsheng Luo
[“Learning Clear Class Separation for Open-Set 3D Detector in Autonomous Vehicle Via Selective Forgetting”](#)
[RO-MAN 2023] *32nd IEEE International Conference on Robot and Human Interactive Communication*
- Weiming Qu, Tianlin Liu, [Weikai Lin](#), Dingsheng Luo
[“A Review of Robot Learning”](#)
[Acta 2023] *Acta Scientiarum Naturalium Universitatis Pekinensis*, 59(6):1069–1086. DOI: 10.13209/j.0479-8023.2023.086
- Shuai Fang, Yaoyao Wei, [Weikai Lin](#), Jianan Zhang, Tianlin Liu, Dingsheng Luo
[“Approaching Sound Object with Sensorimotor Coordination when Sensors Partially Damaged”](#)
[ICDL 2021] *IEEE International Conference on Development and Learning*

PREPRINTS AND WORKSHOPS

- Yaoyao Wei, Shuai Fang, [Weikai Lin](#), Jianan Zhang, Dingsheng Luo
“[Acquiring Robot Navigation Skill with Knowledge Learned from Demonstration](#)”
[ICDL 2021] *IEEE International Conference on Development and Learning*
- Tianyu Hu, [Weikai Lin](#), Weizhi Zhang, Jing Ma, Song Wang
“[MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents](#)”
[arXiv 2026] *arXiv:2605.00356*
- Bo Yan, [Weikai Lin](#), Yada Zhu, Song Wang
“[SafeDream: Safety World Model for Proactive Early Jailbreak Detection](#)”
[arXiv 2026] *arXiv:2604.16824*
- [Weikai Lin*](#), Tianrui Ma*, Adith Bloor, Yu Feng, Ruofan Xing, Xuan Zhang, Yuhao Zhu
“[SnapPix: Efficient-Coding-Inspired In-Sensor Compression for Edge Vision](#)”
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [[link](#)]
- Ruofan Xing, Arman Akbari, [Weikai Lin](#), Adith Bloor, Alexander Montes McNeil, Michael Moebius, Yongmin Liu, Yuhao Zhu, Xuan Zhang
“[HoloCode: Hybrid Optical-Electronic Edge Encoding for Privacy-Preserving Cloud Training](#)”
[CVPR 2026 Workshop] *CVPR 2026 On-Sensor Vision Workshop* [[link](#)]
- Han Yan, Yifei Zou, [Weikai Lin](#), Xuan Zhang, Yuhao Zhu
“[Do Energy-Harvesting Pixels Reduce Carbon Cost of Visual Computing Systems?](#)”
[HotEthics 2026] *Workshop on Ethical Systems and Architecture Design 2026* [[link](#)]
- [Weikai Lin](#), Yu Feng, Yuhao Zhu
“[RTGS: Enabling Real-Time Gaussian Splatting on Mobile Devices Using Efficiency-Guided Pruning and Foveated Rendering](#)”
[arXiv 2024] *arXiv:2407.00435* (preprint and shortened version of MetaSapiens)

PATENTS AND OTHER PROJECTS

- Dingsheng Luo, Xihong Wu, Yifan Yuan, Wenfei Hu, Weiming Qu, Yudi Zou, Jiawen Wang, Hongyu Fang, [Weikai Lin](#)
Control Method and System for Improving Operation Precision of Robot Arm
[Patent] CN202310068025.4, China; published as CN116079730A, 2023
- Dingsheng Luo, Xihong Wu, Shuai Fang, Jianan Zhang, [Weikai Lin](#), Tianlin Liu
Active Auditory Positioning Method for Map-Free Navigation
[Patent] CN202210079214.7, China; published as CN114563011A, 2022
- Dingsheng Luo, Xihong Wu, Jianan Zhang, Shuai Fang, Tianlin Liu, [Weikai Lin](#), Hongyu Fang
Robot Map-Free Navigation Method Based on Time Sequence Information Modeling
[Patent] CN202110018866.5, China; published as CN112857370A, 2021
- “[Exploiting Human Color Discrimination for Memory- and Energy-Efficient Image Encoding in Virtual Reality: An FPGA Demo](#)”
[Open-Source Project] Open-sourced project with optimized CPU/GPU/FPGA implementations. [[code](#)]

TEACHING EXPERIENCE

Teaching Assistant, University of Rochester

- Fall 2025, CSC 259/459 *Computer Imaging and Graphics*
- Spring 2025, CSC 252/452 *Computer Organization*
- Fall 2024, CSC 257/457 *Computer Networks*

EXTERNAL SERVICE

Reviewer

- [SIGGRAPH](#): 2025, 2026
- [SIGGRAPH Asia](#): 2025
- [NeurIPS](#): 2026
- [ISMAR](#): 2026
- [WACV](#): 2026
- [TCAD](#): 2025
- [JCST](#): 2026